

Project Phase 2

Grammar

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Program $\rightarrow$ Functions

Functions $\rightarrow$ Function Functions $| \epsilon$

Function $\rightarrow$ "function" "identifier" ";" "beginparams" Declarations "endparams" "beginlocals" Declarations "endlocals" "beginbody" Statements "endbody"

Declarations $\rightarrow$ Declaration ";" Declarations $| \epsilon$

Declaration $\rightarrow$ Idents ":" "integer" $|$ Idents ":" "enum" "(" Idents ")" $|$ Idents ":" "array" "[" "number" "]" "of" "integer"

Idents $\rightarrow$ Ident $|$ Ident "," Idents

Ident $\rightarrow$ "identifier"

Statements $\rightarrow$ Statement ";" $|$ Statement ";" Statements

Statement $\rightarrow$ StateA $|$ StateB $|$ StateC $|$ StateD $|$ StateE $|$ StateF $|$ StateG $|$ StateH

StateA $\rightarrow$ Var ":"=" Expression

StateB $\rightarrow$ "if" Bool-Exp "then" Statements "endif" $|$ "if" Bool-Exp "then" Statements "else" Statements "endif"

StateC $\rightarrow$ "while" Bool-Exp "beginloop" Statements "endloop"

StateD $\rightarrow$ "do" "beginloop" Statements "endloop" "while" Bool-Exp

StateE $\rightarrow$ "read" Vars

StateF $\rightarrow$ "write" Vars

StateG $\rightarrow$ "continue"

StateH $\rightarrow$ "return" Expression

Bool-Exp $\rightarrow$ Relation-And-Expr $|$ Relation-And-Expr "or" Bool-Exp

Relation-And-Expr $\rightarrow$ Relation-Expr $|$ Relation-Expr "and" Relation-And-Expr

Relation-Expr $\rightarrow$ Rel-Exp $|$ "not" Rel-Exp

Rel-Exp $\rightarrow$ Expression Comp Expression $|$ "true" $|$ "false" $|$ "(" Bool-Exp ")"

Comp $\rightarrow$ "==" $|$ "<>" $|$ "<" $|$ ">" $|$ "<=" $|$ ">="

Expressions $\rightarrow$ Expression $|$ Expression "," Expressions

$\text{Expression} \rightarrow \text{Mult-Expr} \mid \text{Mult-Expr } "+" \text{ Expression} \mid \text{Mult-Expr } "-" \text{ Expression}$
 $\text{Mult-Expr} \rightarrow \text{Term} \mid \text{Term } "*" \text{ Mult-Expr} \mid \text{Term } "/" \text{ Mult-Expr} \mid \text{Term } "%" \text{ Mult-Expr}$
 $\text{Term} \rightarrow \text{TermNum} \mid "-" \text{ TermNum} \mid \text{TermIdent}$
 $\text{TermNum} \rightarrow \text{Var} \mid \text{"number"} \mid "(" \text{ Expression } ")"$
 $\text{TermIdent} \rightarrow \text{"identifier"} "(" \text{ Expressions } ")" \mid \text{"identifier"} "(" ")"$
 $\text{Vars} \rightarrow \text{Var} \mid \text{Var } "," \text{ Vars}$
 $\text{Var} \rightarrow \text{"identifier"} \mid \text{"identifier"} "[" \text{ Expression } "]"$